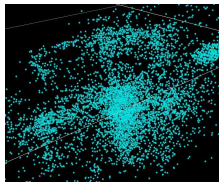
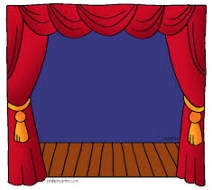
Outline



Properties of language



Distributional semantics



Frame semantics



Model-theoretic semantics



Interactive learning



Reflections

Word meaning revisited

sold

Word meaning revisited

sold

Distributional semantics: all the contexts in which *sold* occurs

...was sold by... *...sold me that piece of...*

- Can find similar words/contexts and generalize (dimensionality reduction), but monolithic (no internal structure on word vectors)

Word meaning revisited

sold

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Frame semantics: meaning given by a **frame**, a stereotypical situation

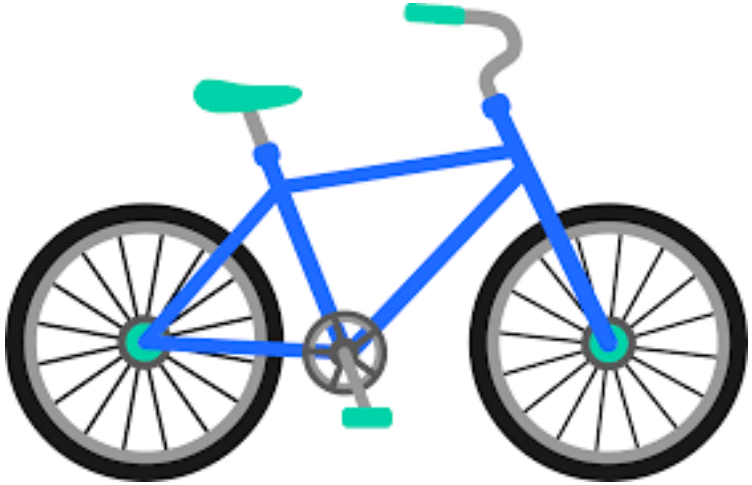


An example

Cynthia sold the bike for $200.

An example

Cynthia sold the bike for $200.



Commercial transaction

SELLER : *Cynthia*

GOODS : *the bike*

PRICE : *$200*

[Fillmore, 1977; Langacker, 1987]

Two properties of frames

Prototypical: don't need to handle all the cases

widow

[Fillmore, 1977; Langacker, 1987]

Two properties of frames

Prototypical: don't need to handle all the cases

widow

- Frame: woman marries one man, man dies

[Fillmore, 1977; Langacker, 1987]

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- Frame: woman marries one man, man dies
- What if a woman has 3 husbands, 2 of which died?

[Fillmore, 1977; Langacker, 1987]

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Profiling: highlight one aspect

- *sell* is seller-centric, *buy* is buyer-centric

Cynthia sold the bike (to Bob).
Bob bought the bike (from Cynthia).

[Fillmore, 1977; Langacker, 1987]

Two properties of frames

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- What if a woman has 3 husbands, 2 of which died?

Profiling: highlight one aspect

- *sell* is seller-centric, *buy* is buyer-centric

Cynthia sold the bike (to Bob).
Bob bought the bike (from Cynthia).

- *rob* highlights person, *steal* highlights goods

Cynthia robbed Bob (of the bike).
Cynthia stole the bike (from Bob).

From syntax to semantics



From syntax to semantics



Many **syntactic alternations** with different arguments/verbs:

Cynthia sold the bike to Bob for $200.

The bike sold for $200.

From syntax to semantics



Many **syntactic alternations** with different arguments/verbs:

Cynthia sold the bike to Bob for $200.

The bike sold for $200.

Bob bought the bike from Cynthia.

The bike was bought by Bob.

The bike was bought for $200.

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From syntax to semantics



Many **syntactic alternations** with different arguments/verbs:

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Goal: **syntactic positions** $\Rightarrow$ **semantic roles**

Semantic role labeling

Task:

Input: *Cynthia sold the bike to Bob for $200*

Semantic role labeling

Task:

Input:	*Cynthia*	*sold*		*the bike*	*to*	*Bob*	*for*	*$200*
Output:	SELLER	**PREDICATE**		GOODS		BUYER		PRICE

Semantic role labeling

Task:

Input:	*Cynthia*	*sold*		*the bike*	*to*	*Bob*	*for*	*$200*
Output:	SELLER	**PREDICATE**		GOODS		BUYER		PRICE

Subtasks:

1. Frame identification (PREDICATE)

2. Argument identification (SELLER, GOODS, etc.)

Types of semantics

Every *non-blue block is next to* **some** *blue block.*

Types of semantics

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Types of semantics

Every *non-blue block is next to* **some** *blue block.*

Distributional semantics: *block* is like *brick*, *some* is like *every*

Frame semantics: *is next to* has two arguments, *block* and *block*

Model-theoretic semantics: tell the difference between





[database]

Executable semantic parsing

What is the largest city in Europe by population?



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

Cities



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

Cities Europe



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

Cities ContainedBy(Europe)



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

Cities ∩ ContainedBy(Europe)



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

Cities ∩ ContainedBy(Europe) Population



[database]

Executable semantic parsing

What is the largest city in Europe by population?

semantic parsing

argmax(Cities ∩ ContainedBy(Europe), Population)



[database]

Executable semantic parsing

What is the largest city in Europe by population?



semantic parsing

argmax(Cities ∩ ContainedBy(Europe), Population)



execute

`Istanbul`



[calendar]

Executable semantic parsing

Remind me to buy milk after my last meeting on Monday.



[calendar]

Executable semantic parsing

Remind me to buy milk after my last meeting on Monday.



semantic parsing

Add(Buy(Milk), argmax(Meetings ∩ HasDate(2016-07-18), EndTime))



[calendar]

Executable semantic parsing

Remind me to buy milk after my last meeting on Monday.

semantic parsing

Add(Buy(Milk), argmax(Meetings ∩ HasDate(2016-07-18), EndTime))

execute

[reminder added]



[context]

Executable semantic parsing

[sentence]

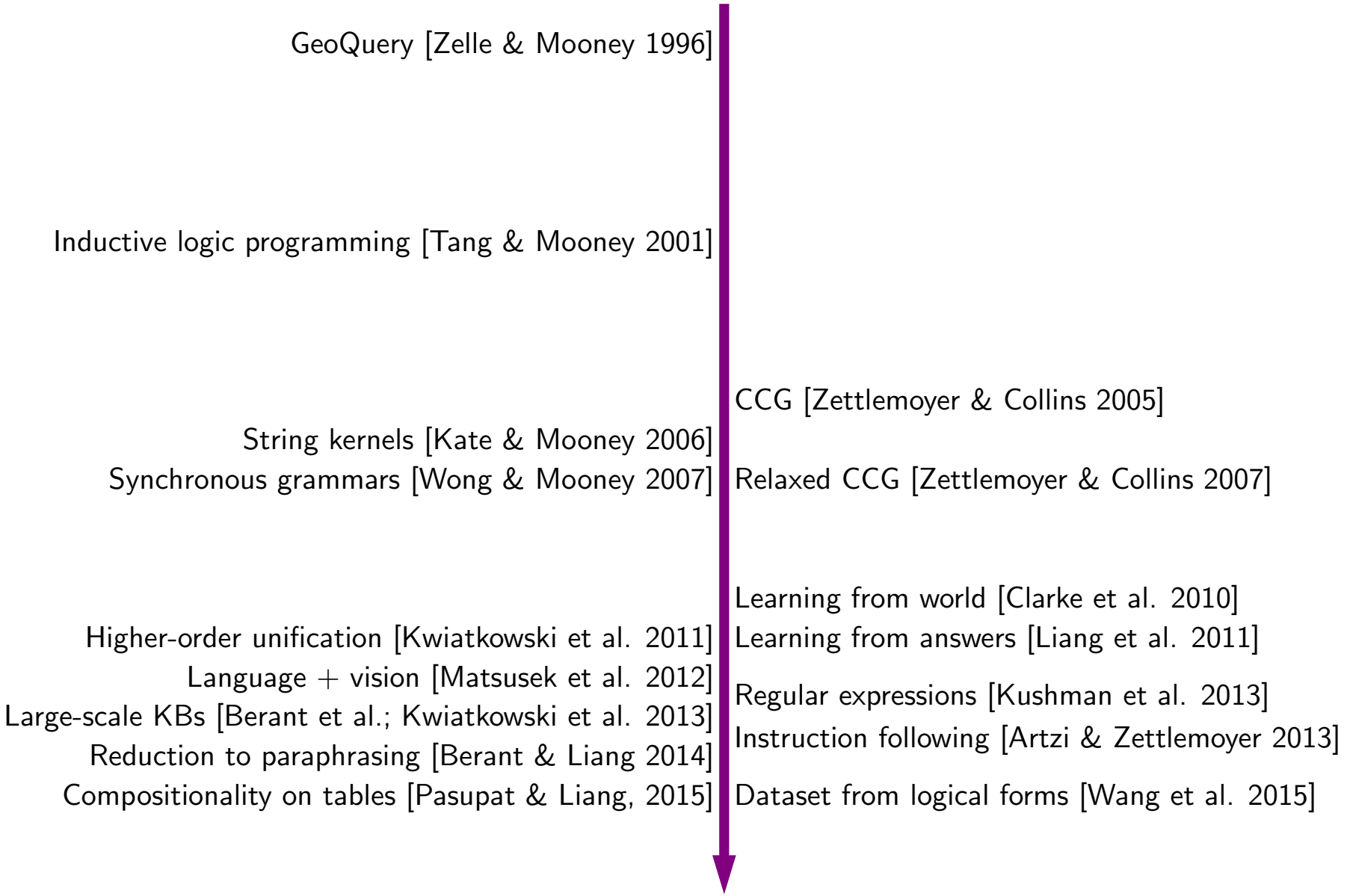
semantic parsing

[program]

execute

[behavior]

A brief history of semantic parsing




Richard Montague

Compositional semantics

cities in Europe



Cities ∩ ContainedBy(Europe)



Richard Montague

Compositional semantics

cities in Europe

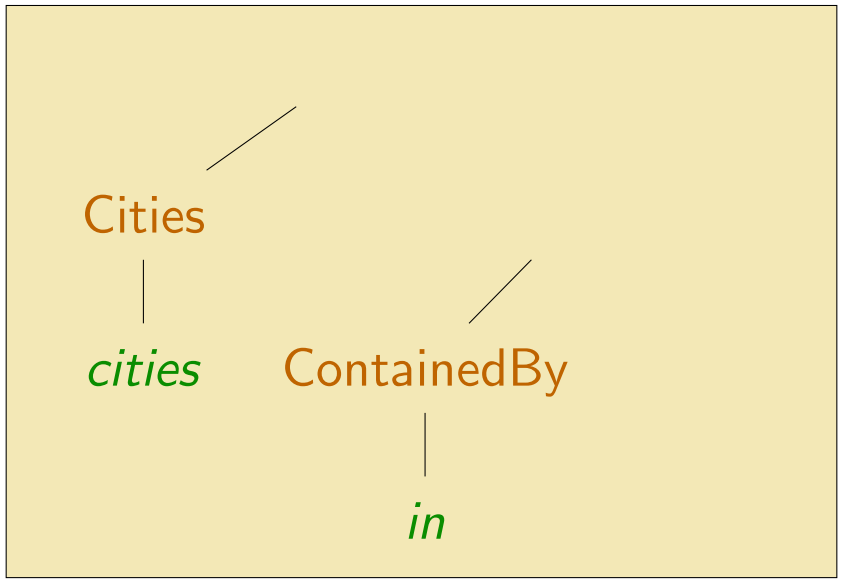


Cities ∩ ContainedBy(Europe)


Richard Montague

Compositional semantics

cities in Europe

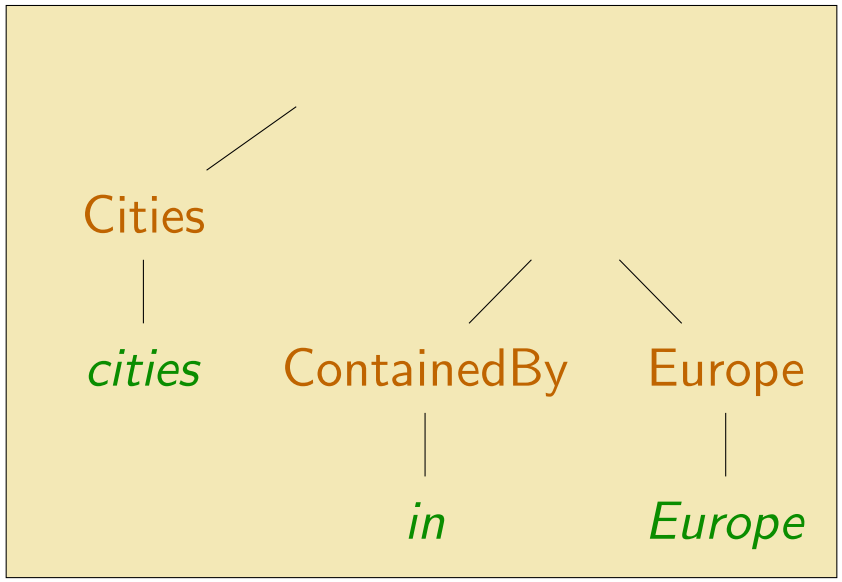


Cities ∩ ContainedBy(Europe)


Richard Montague

Compositional semantics

cities in Europe

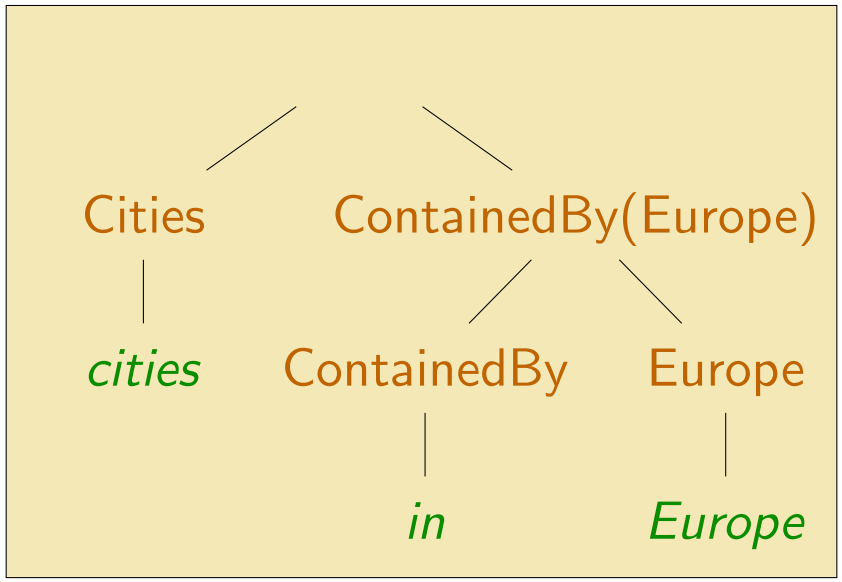


Cities ∩ ContainedBy(Europe)


Richard Montague

Compositional semantics

cities in Europe

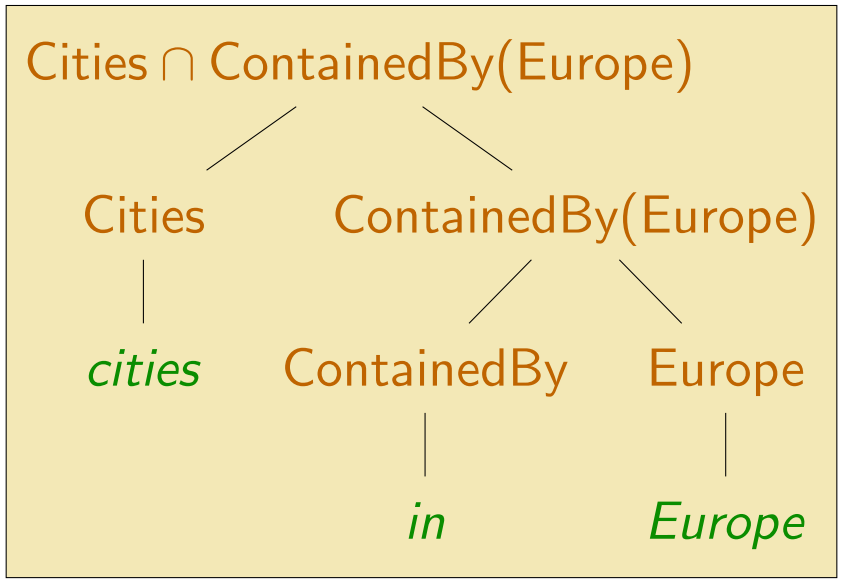


Cities ∩ ContainedBy(Europe)


Richard Montague

Compositional semantics

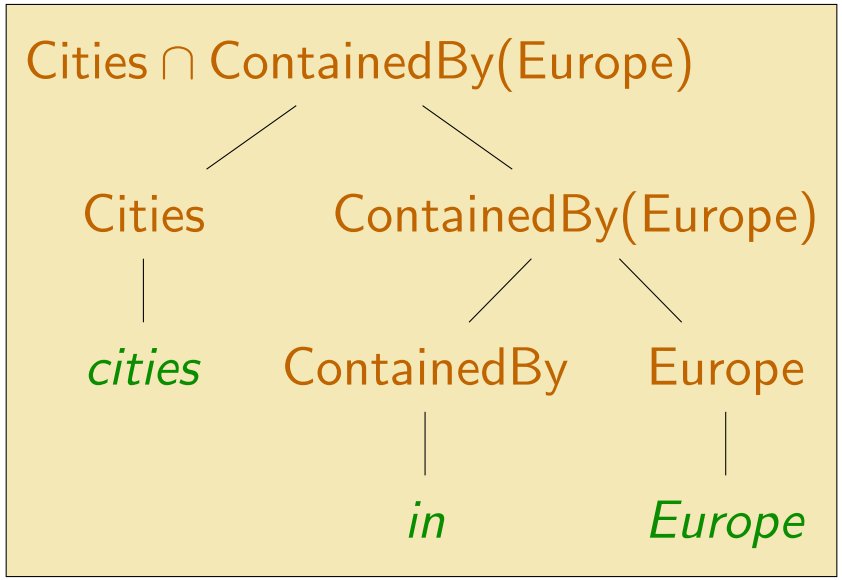
cities in Europe



Cities ∩ ContainedBy(Europe)

Language variation

cities in Europe



Cities ∩ ContainedBy(Europe)

Language variation

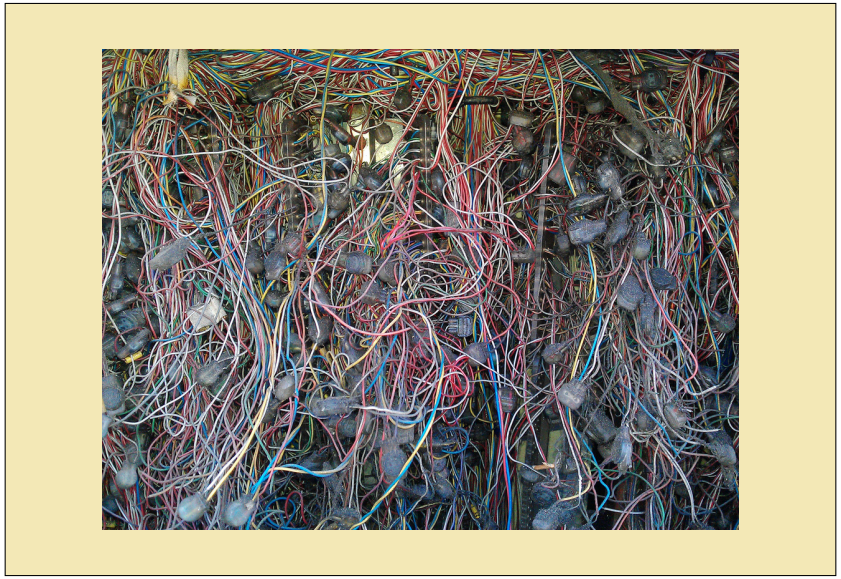
cities in Europe

European cities

cities that are in Europe

cities located in Europe

cities on the European continent



Cities ∩ ContainedBy(Europe)

Language variation

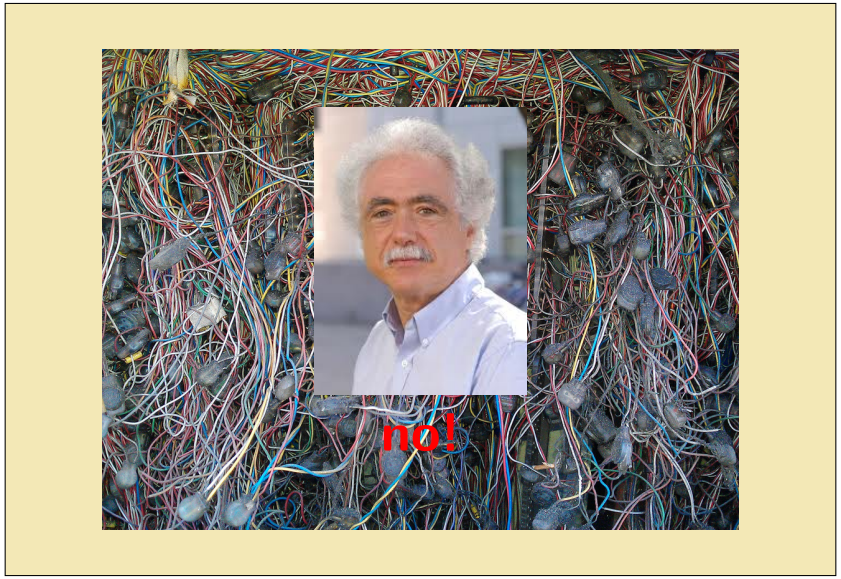
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